

Using (14-99a) and (14-99b)

$$f_1 = \frac{10 \times 10^9}{\sqrt{1 + 1/14.14}} = 9.664 \text{ GHz}$$

$$f_2 = 10 \times 10^9 \sqrt{1 + 1/14.14} = 10.348 \text{ GHz}$$

The relative ratio of the two lengths according to (14-98) is

$$\frac{L}{W} = 1 + \frac{1}{Q_t} = 1 + \frac{1}{14.14} = 1.07$$

which makes the patch nearly square.

14.8 ARRAYS AND FEED NETWORKS

Microstrip antennas are used not only as single elements but are very popular in arrays [17], [23], [30], [31], [50], [51], [54], [63]–[65], and [74]–[77]. As discussed in Chapter 6, arrays are very versatile and are used, among other things, to synthesize a required pattern that cannot be achieved with a single element. In addition, they are used to scan the beam of an antenna system, increase the directivity, and perform various other functions which would be difficult with any one single element. The elements can be fed by a single line, as shown in Figure 14.36(a), or by multiple lines in a feed network arrangement, as shown in Figure 14.36(b). The first is referred to as a *series-feed network* while the second is referred to as a *corporate-feed network*.

The corporate-feed network is used to provide power splits of 2^n (i.e., $n = 2, 4, 8, 16, 32$, etc.). This is accomplished by using either tapered lines, as shown in Figure 14.37(a), to match 100-ohm patch elements to a 50-ohm input, or using quarter-wavelength impedance transformers, as shown in Figure 14.37(b) [3]. The design of single- and multiple-section quarter-wavelength impedance transformers is discussed in Section 9.7.

Series-fed arrays can be conveniently fabricated using photolithography for both the radiating elements and the feed network. However, this technique is limited to arrays with a fixed beam or those which are scanned by varying the frequency, but it can be applied to linear and planar arrays with single or dual polarization. Also any changes

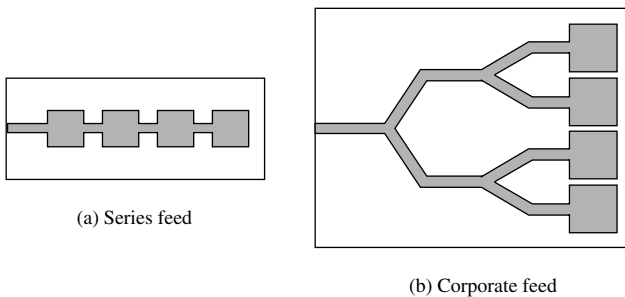


Figure 14.36 Feed arrangements for microstrip patch arrays.

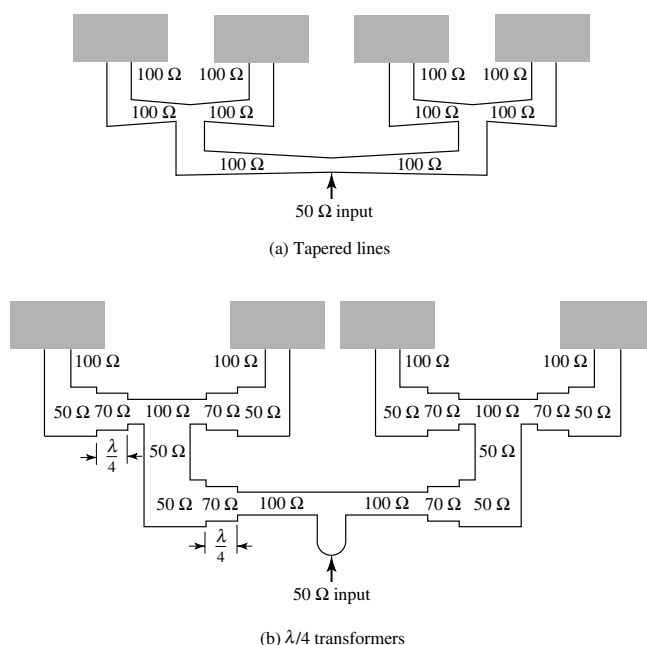


Figure 14.37 Tapered and $\lambda/4$ impedance transformer lines to match 100-ohm patches to a 50-ohm line. (SOURCE: R. E. Munson, "Conformal Microstrip Antennas and Microstrip Phased Arrays," *IEEE Trans. Antennas Propagat.*, Vol. AP-22, No. 1, January 1974. © 1974 IEEE)

in one of the elements or feed lines affects the performance of the others. Therefore in a design it is important to be able to take into account these and other effects, such as mutual coupling, and internal reflections.

Corporate-fed arrays are general and versatile. With this method the designer has more control of the feed of each element (amplitude and phase) and it is ideal for scanning phased arrays, multibeam arrays, or shaped-beam arrays. As discussed in Chapter 6, the phase of each element can be controlled using phase shifters while the amplitude can be adjusted using either amplifiers or attenuators. An electronically-steered phased array (ATDRSS) of 10×10 rectangular microstrip elements, operating in the 2–2.3 GHz frequency range and used for space-to-space communications, is shown in Figure 14.38.

Those who have been designing and testing microstrip arrays indicate that radiation from the feed line, using either a series or corporate-feed network, is a serious problem that limits the cross-polarization and side lobe level of the arrays [38]. Both cross-polarization and side lobe levels can be improved by isolating the feed network from the radiating face of the array. This can be accomplished using either probe feeds or aperture coupling.

Arrays can be analyzed using the theory of Chapter 6. However, such an approach does not take into account mutual coupling effects, which for microstrip patches can be significant. Therefore for more accurate results, full-wave solutions must be performed. In microstrip arrays [63], as in any other array [90], mutual coupling between elements can introduce *scan blindness* which limits, for a certain maximum reflection coefficient, the angular volume over which the arrays can be scanned. For microstrip antennas, this scan limitation is strongly influenced by surface waves within the substrate. This

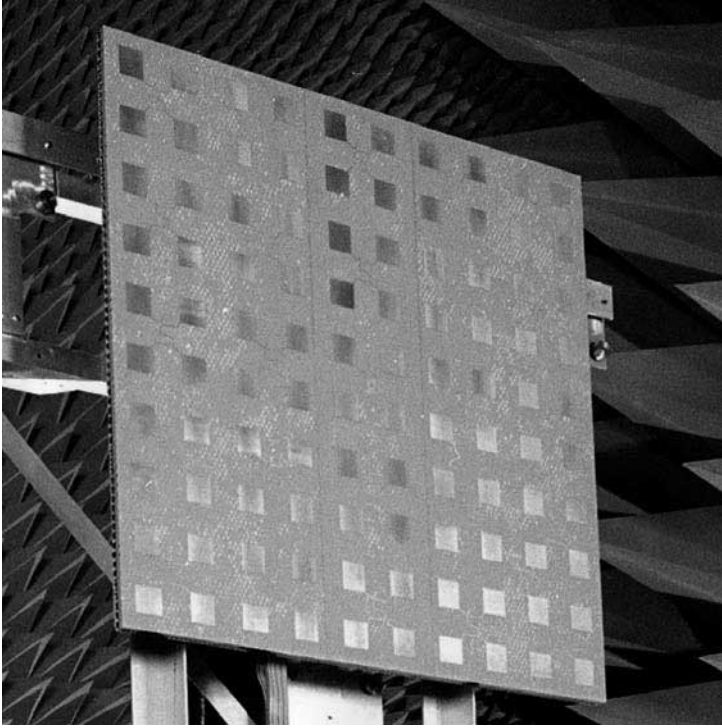


Figure 14.38 Antenna array of 10×10 rectangular microstrip patches, 2–2.3 GHz, for space-to-space communications. (Courtesy: Ball Aerospace & Technologies Corp.)

scan angular volume can be extended by eliminating surface waves. One way to do this is to use cavities in conjunction with microstrip elements [50], [51]. Figure 14.39 shows an array of circular patches backed by either circular or rectangular cavities. It has been shown that the presence of cavities, either circular or rectangular, can have a pronounced enhancement in the E -plane scan volume, especially for thicker substrates [51]. The H -plane scan volume is not strongly enhanced. However the shape of the cavity, circular or rectangular, does not strongly influence the results. Typical results for broadside-matched reflection coefficient infinite array of circular patches, with a substrate $0.08\lambda_0$ thick and backed by circular and rectangular cavities, are shown in Figure 14.40 for the E -plane and H -plane. The broadside-matched reflection coefficient $\Gamma(\theta, \phi)$ is defined as

$$\Gamma(\theta, \phi) = \frac{Z_{in}(\theta, \phi) - Z_{in}(0, 0)}{Z_{in}(\theta, \phi) + Z_{in}^*(0, 0)} \quad (14-103)$$

where $Z_{in}(\theta, \phi)$ is the input impedance when the main beam is scanned toward an angle (θ, ϕ) . The results are compared with those of a conventional cavity (noncavity backed). It is apparent that there is a significant scan enhancement for the E -plane, especially for a VSWR of about 2:1. H -plane enhancement occurs for reflection coefficients greater than about 0.60. For the conventional array, the E -plane response exhibits a large reflection coefficient, which approaches unity, near a scan angle of $\theta_0 = 72.5^\circ$. This is evidence of scan blindness which ideally occurs when the reflection coefficient is unity, and it is attributed to the coupling between the array elements due to leaky